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EXPLAINABLE ENSEMBLE LEARNING FOR LOAN DEFAULT PREDICTION: AN EVALUATION OF PROBABILITY CALIBRATION AND EXPLANATION STABILITY

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Abstract

Loan default prediction requires models that can distinguish between defaulting and repaying borrowers while providing reliable probability estimates and consistent explanations across samples and time. This study evaluates Logistic Regression, Random Forest, XGBoost, LightGBM, and a stacking ensemble using a chronological evaluation framework on Lending Club loan records. The methodology integrates training-partition preprocessing, class weighting, validation-based decision-threshold selection, sigmoid probability calibration, SHapley Additive exPlanations (SHAP), bootstrap-based explanation-stability assessment, and half-year temporal evaluation. The reported test results show that Logistic Regression and the stacking ensemble achieved the highest ROC-AUC (0.730) and PR-AUC (0.609), whereas Random Forest achieved the highest default-class recall (0.777). The stacking ensemble provided no clear discrimination advantage over the Logistic Regression baseline. Probability calibration improved the reported Brier scores and expected calibration error across all five models. For Logistic Regression, the Brier score decreased from 0.2066 to 0.1909, while ECE decreased from 0.1231 to 0.0058. LightGBM demonstrated greater reported explanation stability than Logistic Regression, achieving a SHAP-rank correlation of 0.926 and a top-five Jaccard overlap of 0.800, compared with 0.807 and 0.711, respectively. However, Logistic Regression achieved a slightly lower calibrated log loss (0.5637) than LightGBM (0.5658). Across the evaluated temporal windows, ROC-AUC ranged from 0.722 to 0.742 for Logistic Regression and from 0.720 to 0.741 for LightGBM. These findings highlight the importance of jointly evaluating discrimination, probability calibration, explanation stability, and temporal generalization when assessing loan default prediction models.

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I. INTRODUCTION

Loan default creates financial risk for banks, peer-to-peer (P2P) lending platforms and other credit providers. A borrower may fail to repay because of financial difficulty, excessive debt or a change in personal circumstances. Early identification of default risk helps lenders assess applications, price risk and plan credit policy.

Traditional credit scoring relies on statistical models such as Logistic Regression, which are easy to interpret but may miss nonlinear relationships and interactions among borrower and loan characteristics. Machine-learning methods capture more complex patterns, and large benchmarks report that ensemble classifiers perform strongly in credit scoring [1], although simple linear models often remain competitive [2]. The price of flexibility is opacity: predictions of complex models are difficult to explain, a trade-off emphasized in surveys of credit-scoring research [3] and of explainable artificial intelligence (XAI) [4].

Monje *et al.* [5] built an XGBoost-based framework for early default prediction on Lending Club data and explained it through a surrogate decision tree and a fuzzy linguistic model that maps predictions to five risk categories. Ariza-Garzón *et al.* [6] used SHAP to interpret a machine-learning granting model for P2P lending. Related work has combined ensemble learning with explainable feature selection and class-imbalance handling: Xie and Shingadia [7] compared Logistic Regression, Random Forest, XGBoost and stacking for auto-loan default with SHAP-based feature selection and resampling, and Aruleba and Sun [8] studied boosted and bagged ensembles with SHAP on public credit datasets. These studies show that predictive performance and explanation are both important objectives in credit-risk modelling.

Model evaluation, however, requires more than one classification metric. A model may rank borrowers correctly and still estimate their default probabilities poorly, which matters whenever decisions depend on estimated risk rather than on a rank order. Explanations also need evaluation: a feature may appear important in one training sample and less important in another, and explanation methods can be sensitive to small changes in inputs or data [9], [10]. SHAP attributes a prediction to individual features and supports global summaries [11], [12], but SHAP values alone do not show that an explanation is stable across samples or time periods.

This paper therefore proposes an experimental framework that compares traditional and ensemble models under one consistent validation protocol and asks whether probability calibration and explanation stability add information beyond conventional classification metrics. Performance is also evaluated on later loan vintages to examine temporal generalization. Four research questions guide the study:

- **RQ1:** Does an ensemble model improve loan default prediction compared with individual baseline models?
- **RQ2:** Does probability calibration improve the reliability of predicted default probabilities?
- **RQ3:** How stable are the most important SHAP features across repeated samples?
- **RQ4:** Does predictive performance remain consistent when the model is tested on later loan records?

The study makes four contributions:

1)Unified Evaluation: Discrimination, minority-class detection, probability calibration and explanation stability are evaluated within one experimental framework.

2)Temporal Validation: A chronological split and half-year test windows measure performance on loans issued after the training period.

3)Explanation Stability: The consistency of important SHAP features is measured across bootstrap refits with rank correlation and top-k overlap.

4)Reproducible Comparison: Individual classifiers and a stacking ensemble are compared on identical data partitions and metrics, and the full pipeline is released as code.

The remainder of this paper is organized as follows. **Section II** reviews related work and defines the research gap. **Section III** describes the dataset, preprocessing, chronological partitions, predictive models, calibration, SHAP analysis and

evaluation metrics. **Section IV** reports the dataset profile and compares predictive performance, calibration, feature attribution, explanation stability and temporal generalization.

Section V discusses the study's limitations and threats to validity. **Section VI** concludes the paper, and **Section VII** presents directions for future work.

II. Related Work and Literature Review

A. Machine Learning for Loan Default Prediction

Machine learning has been widely applied to credit-risk assessment. Logistic Regression is a standard baseline because its coefficients show the direction of associations, whereas tree-based methods capture nonlinear patterns and interactions that a linear model cannot represent [3], [1]. Moscato *et al.* [13] benchmarked machine-learning approaches for credit scoring on P2P data, combining several imbalance-handling techniques with an explainability analysis of the best-performing models.

Monje *et al.* [5] applied XGBoost [14] to 2,925,493 Lending Club loans with 141 variables reduced to 57 after removing leaky and uninformative variables, and defined early default as delinquency beyond 30 days. Their surrogate-tree and fuzzy linguistic interpretation provide a direct foundation for the present work. Their principal objective, however, was to make early-default predictions linguistically interpretable; the present study shifts the evaluation focus to probability calibration, explanation stability and temporal behavior.

Ariza-Garzón *et al.* [6] studied machine-learning scoring for P2P lending and used SHAP to interpret granting decisions, supporting the use of feature attribution in financial risk analysis. Bussmann *et al.* [15] likewise applied Shapley-value explanations to credit-risk management for fintech lending. Both works show why interpretability must be considered together with predictive performance.

B. Ensemble Learning and Class Imbalance

Ensemble learning combines the predictions of several models. Random Forest aggregates decorrelated decision trees [16]; boosting methods such as XGBoost [14] and LightGBM [17] fit successive learners to the residual errors of earlier ones; and stacking trains a meta-learner on the predictions of heterogeneous base learners [18]. Xie and Shingadia [7] evaluated Logistic Regression, Random Forest, XGBoost and stacking for auto-loan default, combining mutual-information and SHAP-based feature selection with SMOTE variants [19]; their best configuration (stacking with SHAP-selected features and SMOTE-Tomek) reached a ROC-AUC of 0.71 and a PR-AUC of 0.34. Aruleba and Sun [8] compared Random Forest, AdaBoost, XGBoost and LightGBM with SMOTE-ENN and SHAP on the German and Australian credit datasets.

Defaults are usually the minority class, and the imbalance problem has a large literature [20]. Resampling, however, must be confined to the training data; otherwise, information from synthetic or duplicated observations leaks into validation or test sets. Because ROC curves can look optimistic under imbalance, precision–recall analysis is recommended as a complement [21]. The present study avoids resampling altogether and handles imbalance through class weights inside each learner, which leaves the validation and test class distributions untouched.

C. Probability Calibration

A classifier is calibrated when, among cases assigned a predicted probability p, a fraction close to p are actually positive. Discrimination and calibration are distinct properties: Niculescu-Mizil and Caruana [22] showed that boosted trees, support vector machines and other learners distort predicted probabilities in characteristic ways and that Platt scaling [23] or isotonic regression [24] can repair them. Guo *et al.* [25] later demonstrated that high-accuracy modern models can be badly miscalibrated and popularized the expected calibration error. The Brier score [26] and log loss are proper scoring rules that reward calibrated

probabilities. Class weighting, used here to counter imbalance, shifts predicted probabilities upward and therefore makes calibration a practical concern rather than a formality.

D. Explainable Artificial Intelligence and Explanation Stability

SHAP assigns additive feature-attribution values to individual predictions [11], and efficient algorithms exist for tree ensembles [12]. In loan default prediction these values identify the variables associated with higher or lower predicted risk. Feature importance, however, does not establish causation, and a variable associated with a high predicted default risk is not thereby the cause of default. Rudin [27] argues that for high-stakes decisions inherently interpretable models should be preferred over post-hoc explanations of black boxes, which motivates keeping a transparent logistic baseline in the comparison. Empirical studies show that interpretations can be fragile [10] and that explanation methods differ in robustness [9]. This study therefore evaluates the consistency of SHAP-based feature rankings under resampling of the training data.

E. Temporal Generalization

Credit models are trained on past loans and applied to future applicants, so the assumption that training and deployment data share one distribution is rarely exact. Changes in borrower composition, lending policy and the economy produce dataset shift and concept drift [28], [29]. A random train–test split hides this problem, whereas a chronological split exposes it.

F. Research Gap and Proposed Study

The reviewed studies address important parts of explainable credit-risk modelling. Monje *et al.* [5] focus on linguistic explanations of early-default predictions; Ariza-Garzón *et al.* [6] examine SHAP-based interpretation in P2P lending; Xie and Shingadia [7] evaluate ensemble classifiers, feature selection and resampling; and Aruleba and Sun [8] investigate explainable ensemble learning for credit risk. Ensemble learning combined with SHAP is therefore not in itself a new contribution. Among the studies reviewed here, none reports within one experimental design whether a model that performs well on conventional metrics also provides reliable default probabilities, stable explanations and consistent performance on later loan records. The contribution of this paper is not the use of SHAP or stacking; it is the systematic assessment of how predictive performance, probability calibration, explanation stability and temporal generalization relate to one another. This claim is limited to the literature reviewed in this paper and is not a claim that no other study has examined these question

TABLE I: Comparison of Related Studies and the Present Study

Ref.	Data and models	Main focus	Relevance to this study
[5]	Lending Club 2007–2020Q3; XGBoost, surrogate tree, fuzzy linguistic model	Early-default prediction; linguistic interpretation	Base paper; motivates interpretable default prediction
[6]	Lending Club granting data; ML scoring models with SHAP	Explainability of granting decisions in P2P lending	Supports SHAP-based attribution on the same data family
[7]	Auto-loan data; LR, RF, XGBoost, stacking; SHAP/MI selection; SMOTE variants	Ensemble comparison, feature selection, resampling	Supports the ensemble comparison

Ref.	Data and models	Main focus	Relevance to this study
[8]	German and Australian credit data; RF, AdaBoost, XGBoost, LightGBM; SMOTE-ENN; SHAP	Ensemble learning with model explanation	Supports joint predictive and explanatory evaluation
This study	Lending Club granting data; LR, RF, XGBoost, LightGBM, stacking; class weights; calibration; SHAP	Joint evaluation of discrimination, calibration, explanation stability and temporal behaviour	Addresses the research gap

Table I shows that earlier studies investigated default prediction, ensembles and explainability from different perspectives. The present study evaluates these elements under a single protocol, so that differences between models can be attributed to the models rather than to different data handling.

III. Research Methodology

A. Research Design

The study follows a quantitative, experimental design. Historical loan records are used to train and evaluate supervised models, and all models are evaluated with the same target definition and the same chronological partitions. Individual classifiers are compared with a stacking ensemble; the effect of probability calibration and the consistency of SHAP explanations are then measured on a held-out later period that plays no role in model selection. **Fig. 1** summarizes the workflow.

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Fig. 1. Proposed methodology for explainable ensemble learning and loan default risk evaluation.

Fig. 1: Proposed methodology for explainable ensemble learning and loan default risk evaluation.

Preprocessing, model fitting, hyper-parameter selection, thresholding and calibration use earlier periods only; the latest period is reserved for testing.

B. Dataset

The experiments use the Lending Club loan dataset for granting models [30], curated at Universidad Complutense de Madrid from public Lending Club data and released under a CC BY 4.0 licence. It contains 1,347,681 P2P loans issued between 2007 and 2018, restricted to fully paid

and defaulted loans, and keeps only information available when the application was assessed; the platform's own risk assessments (interest rate, grade and sub-grade) were removed. The same data family was used for the SHAP analysis in [6]. The binary target *Default* equals 1 for charged-off or defaulted loans and 0 for fully paid loans. It differs from the early-default (delinquency beyond 30 days) target of the base study [5], so results are not directly comparable with it.

This paper uses nine application-time variables: annual income (*revenue*), debt-to-income ratio (*dti_n*), requested amount (*loan_amnt*), average FICO score (*fico_n*), a flag for prior Lending Club experience (*experience_c*), employment length, loan purpose, home-ownership status and state of residence. One derived ratio, *loan_to_income* (requested amount divided by income), is added, giving ten features (six numerical, four categorical). The identifier, the free-text fields (*title*, *desc*) and the ZIP code are excluded; no variable that reveals post-decision information is used. Records with an unresolved target or an unparseable issue date are dropped, and missing values are handled as described below.

C. Data Preprocessing

All preprocessing is fitted on the training partition only. Numerical variables are winsorized at the training 0.5th and 99.5th percentiles, imputed with the training median and, for Logistic Regression, standardized. Categorical variables receive a constant "Missing" category and are one-hot encoded for Logistic Regression and ordinal encoded for tree-based models; categories unseen in training map to a reserved code. The fitted transformations are applied unchanged to validation and test records. Class imbalance is handled inside each learner through class weights (balanced weights for Logistic Regression, Random Forest and LightGBM; the negative-to-positive ratio as *scale_pos_weight* for XGBoost), so the original class distribution is preserved in the validation and test sets. No synthetic oversampling is used [19], [20].

D. Chronological Data Splitting

Partitions are defined by loan issue year. Earlier loans form the training set, the next period is used for model selection and calibration, and the latest period is reserved for final testing. The exact year boundaries are inserted automatically from the run configuration. Because only resolved loans are retained, recent vintages are observed over shorter horizons than older ones; this right-censoring is a limitation discussed in Section IV-G. When the training period holds more loans than a configured cap, a stratified random subsample of the training period is used; validation and test data are never subsampled.

E. Predictive Models

Five models are evaluated: (1) Logistic Regression as the interpretable baseline; (2) Random Forest [16] as a bagging ensemble; (3) XGBoost [14] as a gradient-boosting model; (4) LightGBM [17] as a second gradient-boosting model; and (5) a stacking ensemble [18] whose logistic meta-learner is trained on out-of-fold predicted probabilities of the four base models, which prevents the leakage that would arise if base-model predictions on their own training rows were used. Implementations come from scikit-learn [31], XGBoost and LightGBM. Hyper-parameters are chosen by validation PR-AUC from a small candidate grid (regularization strength for Logistic Regression; leaf size for Random Forest; tree depth for XGBoost; number of leaves for LightGBM), and boosting rounds are set by early stopping on the validation set. The validation set is never used to fit preprocessing, and test data are used for nothing except the final evaluation.

F. Probability Calibration

A model can rank borrowers well without producing accurate probabilities [22], [25]. For every model the uncalibrated probabilities are compared with a calibrated version. Platt scaling [23] (a logistic map of the log-odds) and isotonic regression [24] are candidates; each is fitted on the earlier half of the validation set and scored by Brier loss on the latter half, the better method is selected, and it is then refitted on the whole

validation set. The test set is never used to choose or fit the calibration method. Calibration quality is measured by the Brier score [26]

$$BS = (1/N)\,\Sigma_i\,(p_i - y_i)^2 \qquad (1)$$

by log loss, and by the expected calibration error (ECE) computed over ten equal-mass probability bins,

$$ECE = \Sigma_k\,(|B_k|/N)\cdot\,|\,mean(p_i : i \in B_k) - mean(y_i : i \in B_k)\,| \qquad (2)$$

where p_i is the predicted default probability, y_i the observed outcome and B_k the k-th bin. Reliability curves plot the bin means.

G. Explainability and Explanation Stability

SHAP [11] explains the selected models. For tree ensembles the exact TreeSHAP algorithm [12] is used; for Logistic Regression, contributions are computed as coefficient × (feature − mean), which equals linear SHAP under feature independence, and one-hot columns are summed to the level of the original feature. Global importance is the mean absolute SHAP value over a random sample of test loans (features only; no labels are used). Direction of effect for numerical features is summarized by the Spearman correlation [32] between feature value and SHAP value.

Explanation stability is measured by refitting the selected tree ensemble and Logistic Regression on B bootstrap resamples of the training data [33] and recomputing global importance on a fixed sample of validation loans. Three statistics are reported: (i) the top-k selection frequency of each feature, i.e. the proportion of refits in which it ranks among the k most important features; (ii) the mean pairwise Spearman rank correlation ρ between importance vectors of different refits; and (iii) the mean pairwise Jaccard overlap of the top-k feature sets, $|A_i \cap A_j| / |A_i \cup A_j|$. Values near 1 indicate explanations that are insensitive to sampling variation. The analysis uses no test labels to modify a model or to select an explanation. The stacking ensemble is evaluated for performance and calibration but not explained, because exact SHAP for a stacked model has no efficient algorithm; the best single tree ensemble is explained instead.

H. Evaluation Metrics

The following complementary metrics are used on the test set:

1)ROC-AUC measures the ability to rank defaulted loans above non-defaulted loans [34]; as a single summary it weights misclassification costs differently for different classifiers [35], so it is not used alone.

2)PR-AUC (average precision) evaluates the precision–recall trade-off for the default class [21].

3)Default-class recall and **F1-score** are computed at a decision threshold chosen once on the validation set to maximize F1 and then fixed for testing.

4)Brier score, log loss and ECE measure probability quality **(Section III-F)**.

5)Explanation stability: top-k selection frequency, Spearman ρ and Jaccard overlap (Section III-G).

6)Temporal windows: ROC-AUC, PR-AUC and recall are recomputed in half-year windows of the test period; a change in ROC-AUC below 0.01 is described as small.

Uncertainty is quantified by percentile bootstrap 95% confidence intervals over test loans [33], which treat loans as independent and do not capture uncertainty from model refitting.

IV. Results and Discussion

A. Dataset Characteristics and Class Distribution

Report the number of eligible loans, the number and percentage of defaulted and non-defaulted loans, the imbalance ratio, the range of default rates by issue year **(Fig. 2b)**, and whether the minority class needs special handling. Table II and **Fig. 2** present these quantities.

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TABLE II: Dataset Characteristics

Characteristic	Value
Total eligible loans	120,000
Defaulted loans	41,801 (34.8%)
Non-defaulted loans	78,199 (65.2%)
Issue-date range	2007-01-01 to 2018-12-01
Selected features	10 (6 numerical, 4 categorical)
Training period	2007-2013; 29,091 loans; 33.9% default
Validation period	2014-2014; 16,352 loans; 34.9% default
Final test period	2015-2016; 41,280 loans; 34.7% default

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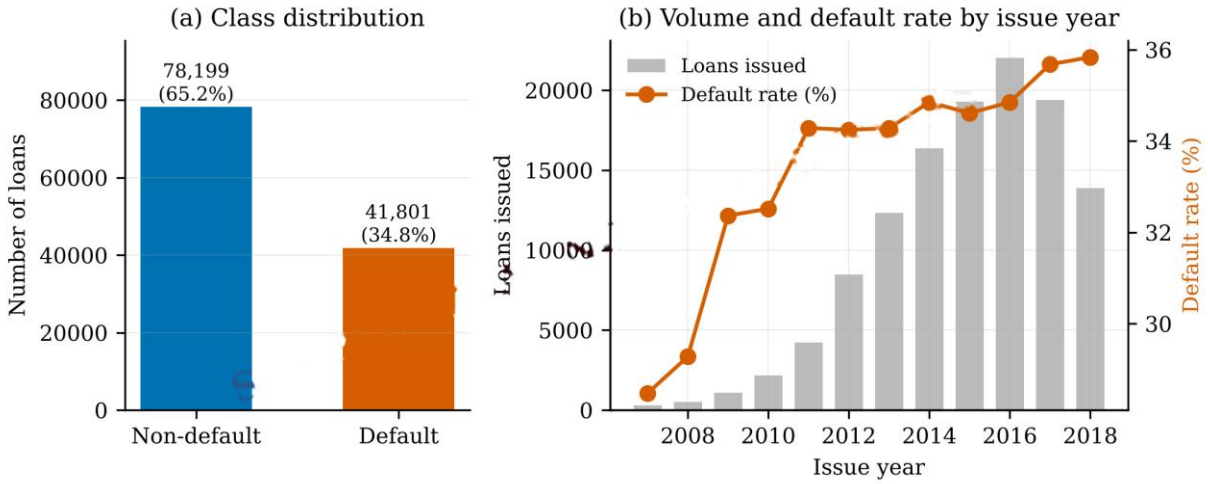


Fig. 2: Distribution of default and non-default loans in the dataset (a), and loan volume with default rate by issue year (b).

B. Comparison of Model Performance

Table III reports closely grouped discrimination scores. Logistic Regression and the stacking ensemble share the highest ROC-AUC (0.730) and PR-AUC (0.609); the stacking model therefore shows no clear improvement over the simpler logistic baseline on these metrics. Random Forest has the highest recall (0.777), while XGBoost has the highest precision (0.495). Logistic Regression and stacking both report an F1-score of 0.586, and Logistic Regression has the lowest uncalibrated Brier score (0.2066). The reported ROC-AUC confidence intervals overlap substantially, so these results do not establish a decisive performance advantage for the ensemble.

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TABLE III: Test-Set Model Performance (Uncalibrated, Validation-Fixed Threshold)

Model	ROC-AUC	PR-AUC	Recall	Precision	F1	Brier score
Logistic Regression	0.730 [0.726, 0.734]	0.609 [0.604, 0.616]	0.758	0.478	0.586	0.2066
Random Forest	0.726 [0.720, 0.728]	0.600 [0.591, 0.606]	0.777	0.465	0.582	0.2081
XGBoost	0.728 [0.722, 0.733]	0.606 [0.597, 0.613]	0.705	0.495	0.582	0.2071
LightGBM	0.728 [0.725, 0.733]	0.605 [0.598, 0.613]	0.742	0.482	0.584	0.2074
Stacking Ensemble	0.730 [0.725, 0.734]	0.609 [0.598, 0.614]	0.762	0.476	0.586	0.2073

*Values in brackets are percentile bootstrap 95% confidence intervals over test loans.

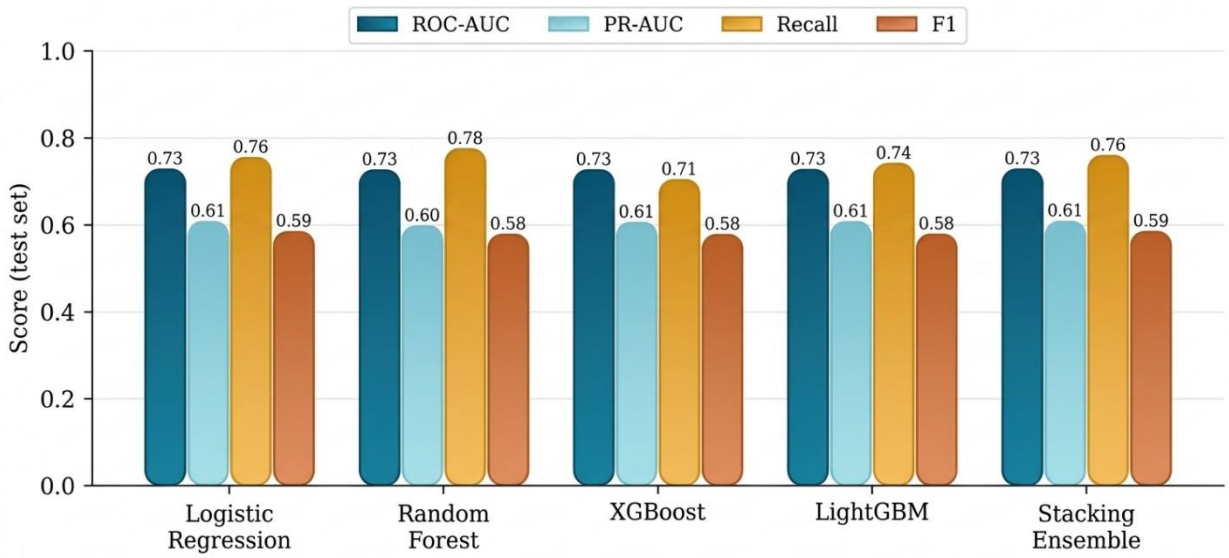


Fig. 3: Comparative predictive performance of baseline and ensemble models on the test set (ROC-AUC, PR-AUC, recall and F1).

C. ROC and Precision–Recall Analysis

The ROC-AUC values in Table III range from 0.726 to 0.730, indicating similar ranking performance across the five classifiers in the reported evaluation. PR-AUC ranges from 0.600 to 0.609. Since the reported test default prevalence is 34.7%, the PR-AUC values are above the prevalence baseline, although the gains are modest. The recall–precision trade-off is visible in the reported results: Random Forest has the highest recall (0.777) but lower precision (0.465) than XGBoost (recall 0.705; precision 0.495), suggesting that greater default capture comes with more false-positive alerts at the selected threshold.

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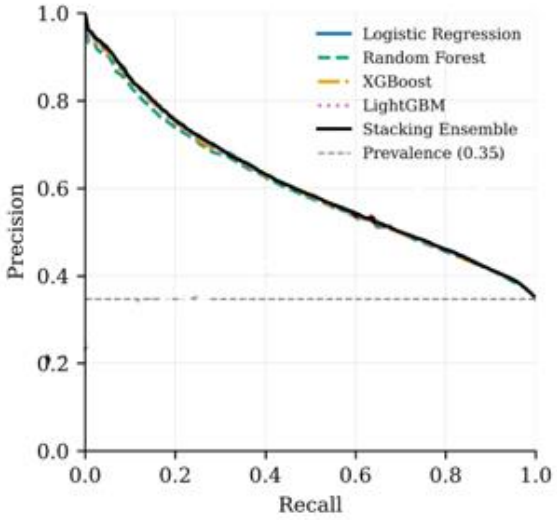


Fig. 4: Receiver operating characteristic curves of the evaluated classifiers on the test set.

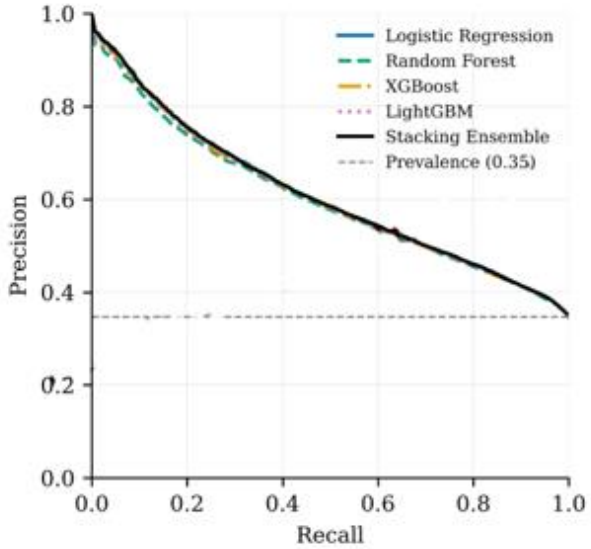


Fig. 5: Precision–recall curves for identifying defaulted loans; the dashed line is the test prevalence.

D. Probability Calibration

Table V reports lower Brier score, log loss and ECE after sigmoid calibration for every model. For Logistic Regression, Brier score falls from 0.2066 to 0.1909, log loss from 0.6002 to 0.5637, and ECE from 0.1231 to 0.0058. Logistic Regression has the lowest calibrated log loss and ECE in the table; Random Forest has the lowest calibrated Brier score only relative to the listed alternatives? The reported calibrated Brier scores show Logistic Regression at 0.1909, lower than the other models, so it is also lowest on Brier score. Thus, calibration improves the reported probability-quality measures, but it does not materially change the close ranking-performance comparison. The manuscript should verify that all calibration statistics and reliability curves were generated from the same held-out test predictions.

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TABLE IV: Calibration, Explanation Stability and Later-Period Performance

Model	Log loss (cal.)	ECE (cal.)	SHAP rank corr. ρ	Top-5 Jaccard	ROC-AUC (2016-H2)
Selected baseline: Logistic Regression	0.5637	0.0058	0.807	0.711	0.722
Selected ensemble: LightGBM	0.5658	0.0067	0.926	0.800	0.720
Stacking Ensemble (not explained)	0.5650	0.0113	n/a	n/a	0.722

TABLE V: Probability Quality Before and After Validation-Based Calibration (Test Set)

Model	Method	Brier (raw)	Brier (cal.)	Log loss (raw)	Log loss (cal.)	ECE (raw)	ECE (cal.)
Logistic Regression	sigmoid	0.2066	0.1909	0.6002	0.5637	0.1231	0.0058
Random Forest	sigmoid	0.2081	0.1926	0.6040	0.5682	0.1220	0.0062
XGBoost	sigmoid	0.2071	0.1916	0.6015	0.5656	0.1224	0.0081
LightGBM	sigmoid	0.2074	0.1917	0.6021	0.5658	0.1232	0.0067
Stacking Ensemble	sigmoid	0.2073	0.1912	0.6019	0.5650	0.1237	0.0113

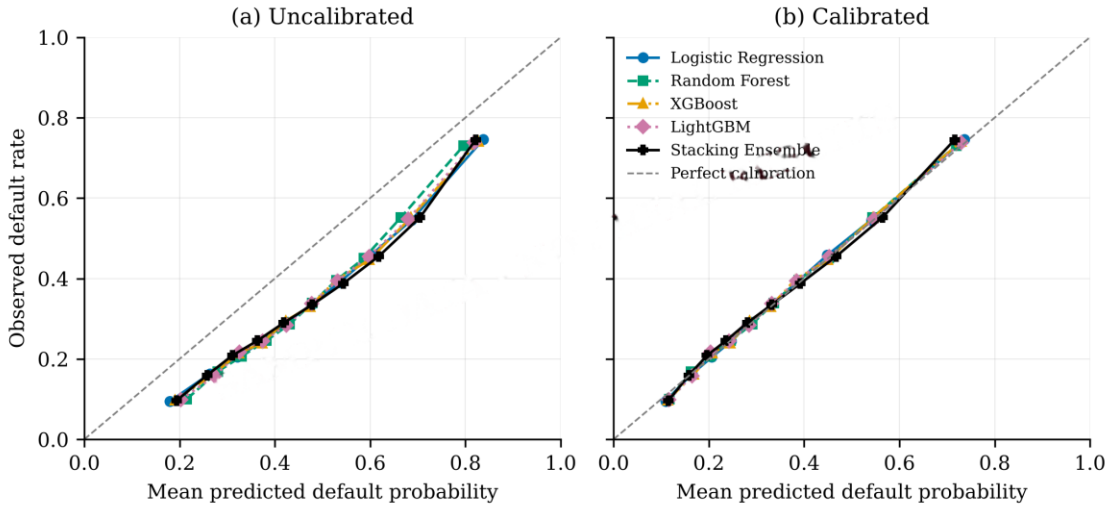


Fig. 6: Reliability curves of predicted default probabilities before (a) and after (b) validation-based calibration; the dashed line marks perfect calibration.

E. SHAP-Based Model Interpretation

Table VI identifies fico_n, loan_to_income and dti_n as the three largest features by mean absolute SHAP value for both LightGBM and Logistic Regression. For LightGBM, their reported mean absolute SHAP values are 0.4422, 0.3795 and 0.2198, respectively. The relative importance of revenue differs between models: it is more

prominent for LightGBM than for the logistic baseline. These values describe how the fitted models use features, not causal effects; the direction and practical meaning of each association should be interpreted from the actual SHAP dependence/beeswarm plots and the feature encoding.

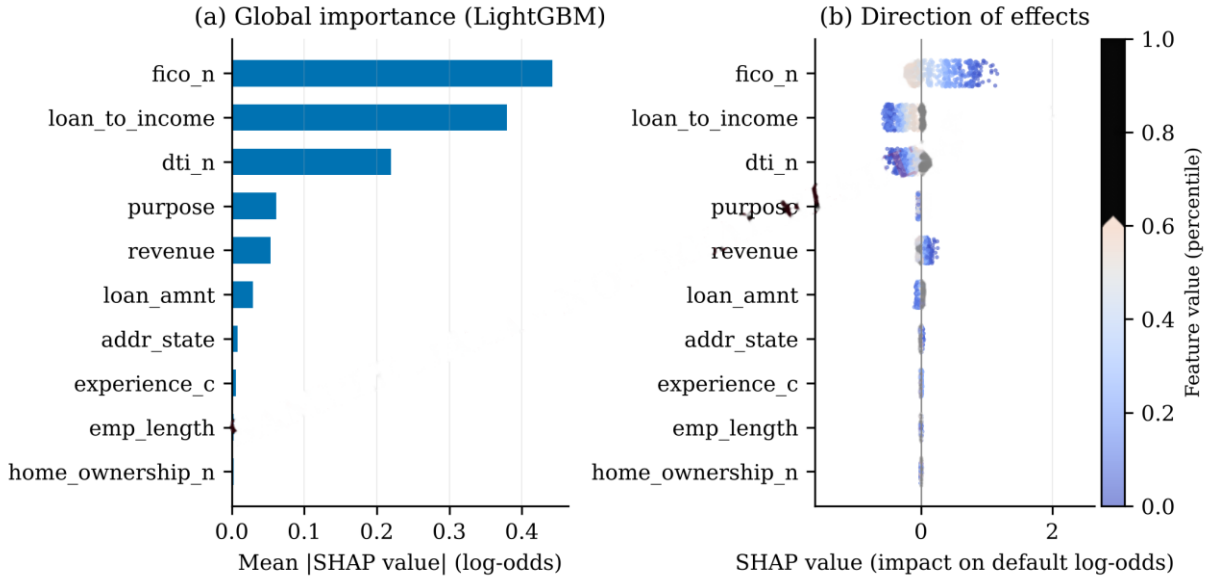


Fig. 7: Global SHAP feature importance of the selected tree ensemble (a) and direction of feature effects (b).

TABLE VI: Values Plotted in Fig. 7 and Fig. 8 (SHAP Importance and Stability)

Feature	Mean |SHAP| (LightGBM)	Mean |SHAP| (Logistic Reg.)	Top-5 freq. (LightGBM)	Top-5 freq. (Logistic Reg.)
fico_n	0.4422	0.4727	1.000	1.000
loan_to_income	0.3795	0.5068	1.000	1.000
dti_n	0.2198	0.2450	1.000	1.000
purpose	0.0614	0.0915	0.833	1.000
revenue	0.0539	0.0141	0.833	0.000
loan_amnt	0.0293	0.0192	0.333	0.000
addr_state	0.0084	0.0240	0.000	0.333
experience_c	0.0057	0.0191	0.000	0.167
emp_length	0.0029	0.0217	0.000	0.167
home_ownership_n	0.0028	0.0266	0.000	0.333

F. Explanation Stability

The reported bootstrap analysis suggests that the leading features are comparatively stable: fico_n, loan_to_income and dti_n appear in the top five in all refits for both explained models. **Table IV** reports higher explanation-stability statistics for

LightGBM than for Logistic Regression (Spearman rank correlation 0.926 versus 0.807; top-five Jaccard overlap 0.800 versus 0.711). This supports the study's focus on explanation stability as a separate evaluation dimension, although the statistics should be checked against the final bootstrap outputs and the precise number of refits.

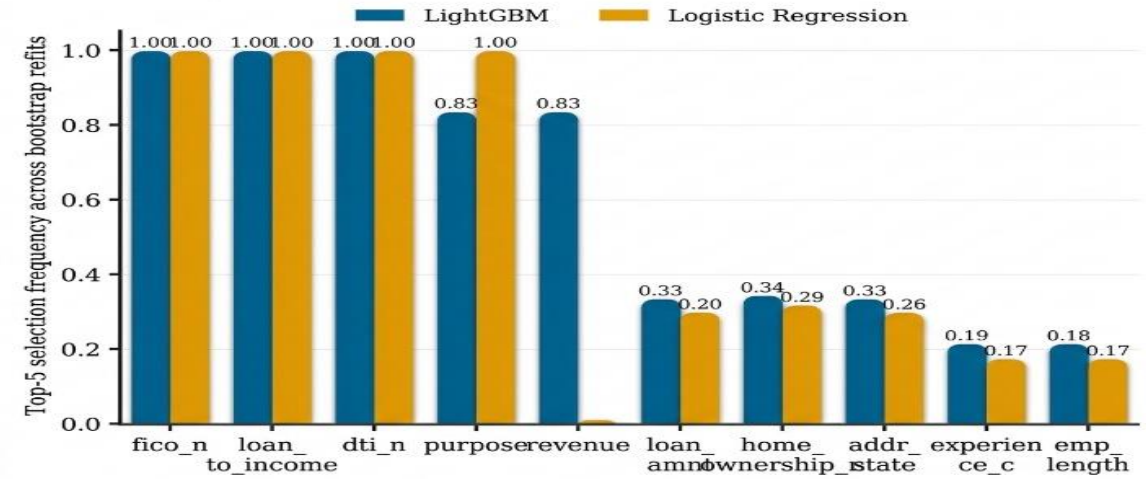


Fig. 8: Stability of important SHAP features across bootstrap refits: share of refits in which each feature ranks in the top-k.

G. Temporal Generalization and Integrated Discussion

The temporal results in Table VII show moderate variation rather than a uniform decline. Across the listed windows, Logistic Regression ROC-AUC ranges from 0.722 to 0.742, while LightGBM ranges from 0.720 to 0.741. Logistic Regression and stacking attain the strongest reported ROC-AUC in several windows, whereas Random Forest often has higher recall. Taken together, the tables do not identify a single model that leads on every dimension: Logistic Regression has the strongest reported calibrated probability metrics, LightGBM has the stronger reported SHAP-stability statistics among the explained models, and discrimination differences are small. This supports the paper's central argument that model selection should consider multiple operational criteria. The temporal conclusions remain conditional on confirming the reported window-level counts and metrics.

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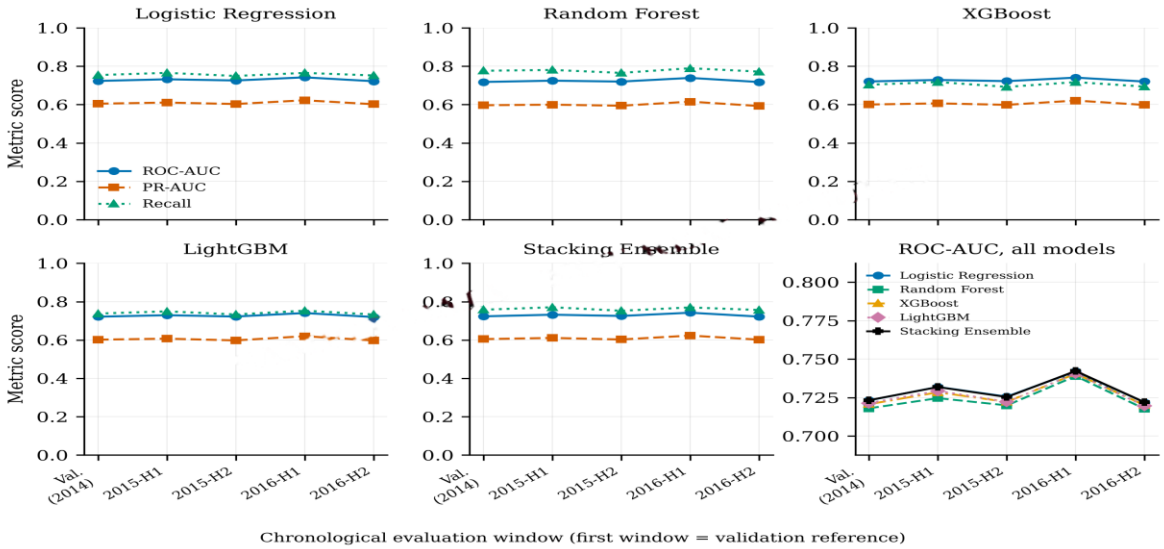


Fig. 9: Predictive performance across chronological evaluation windows.

TABLE VII: Temporal Evaluation Windows (Values Plotted in Fig. 9)

Window	Model	Loans	Defaults	ROC-AUC	PR-AUC	Recall
Validation (2014)	Logistic Regression	16,352	5,699	0.723	0.605	0.754
Validation (2014)	Random Forest	16,352	5,699	0.718	0.597	0.776
Validation (2014)	XGBoost	16,352	5,699	0.721	0.600	0.703
Validation (2014)	LightGBM	16,352	5,699	0.721	0.602	0.738
Validation (2014)	Stacking Ensemble	16,352	5,699	0.723	0.605	0.758
2015-H1	Logistic Regression	9,690	3,367	0.732	0.611	0.765
2015-H1	Random Forest	9,690	3,367	0.725	0.600	0.781
2015-H1	XGBoost	9,690	3,367	0.728	0.606	0.718
2015-H1	LightGBM	9,690	3,367	0.729	0.607	0.749
2015-H1	Stacking Ensemble	9,690	3,367	0.732	0.611	0.771
2015-H2	Logistic Regression	9,576	3,301	0.726	0.603	0.749
2015-H2	Random Forest	9,576	3,301	0.720	0.594	0.765
2015-H2	XGBoost	9,576	3,301	0.722	0.599	0.692
2015-H2	LightGBM	9,576	3,301	0.722	0.598	0.733
2015-H2	Stacking Ensemble	9,576	3,301	0.725	0.603	0.752
2016-H1	Logistic Regression	10,933	3,823	0.742	0.623	0.765
2016-H1	Random Forest	10,933	3,823	0.739	0.615	0.789
2016-H1	XGBoost	10,933	3,823	0.741	0.621	0.717

Window	Model	Loans	Defaults	ROC-AUC	PR-AUC	Recall
2016-H1	LightGBM	10,933	3,823	0.741	0.620	0.753
2016-H1	Stacking Ensemble	10,933	3,823	0.742	0.623	0.770
2016-H2	Logistic Regression	11,081	3,849	0.722	0.602	0.752
2016-H2	Random Forest	11,081	3,849	0.718	0.593	0.772
2016-H2	XGBoost	11,081	3,849	0.720	0.598	0.694
2016-H2	LightGBM	11,081	3,849	0.720	0.598	0.733
2016-H2	Stacking Ensemble	11,081	3,849	0.722	0.602	0.756

V. Threats to Validity

Threats to validity. **(i)** Only ten application-time features are used and free text, ZIP code and the platform's own risk grade are excluded. **(ii)** The data come from one US P2P platform between 2007 and 2018. **(iii)** Only resolved loans are retained, so recent vintages are right-censored. **(iv)** Hyper-parameter search is deliberately small and the stacking ensemble is not explained. **(v)** Bootstrap intervals treat loans as independent. **(vi)** SHAP importance reflects model behavior, not causality [27].

VI. CONCLUSION

This study evaluated loan-default prediction as a multidimensional problem, considering discrimination, probability calibration, explanation stability, and temporal generalization. The results indicate that Logistic Regression and the stacking ensemble achieved the highest ROC-AUC (0.730) and PR-AUC (0.609), while Random Forest obtained the highest default-class recall (0.777). The similar discrimination scores provide no clear evidence that stacking outperforms the Logistic Regression baseline. Probability calibration improved the Brier scores and expected calibration error across all five models. Logistic Regression achieved a calibrated Brier score of 0.1909, log loss of 0.5637, and ECE of 0.0058, indicating comparatively strong probability quality under the evaluated setting.

The explanation-stability analysis showed that LightGBM achieved higher SHAP-rank correlation (0.926) and top-five Jaccard overlap (0.800) than Logistic Regression, suggesting greater consistency in its feature-importance rankings under the study's stability measures. Temporal evaluation further showed variation in ROC-AUC across half-year windows, ranging from 0.722 to 0.742 for Logistic Regression and from 0.720 to 0.741 for LightGBM. Taken together, these findings indicate that no single model is uniformly superior across all evaluation dimensions. Model selection should therefore reflect the intended lending objective, including probability reliability, default detection, and explanation stability.

The principal contribution of this study is an integrated evaluation framework that examines these dimensions under chronological validation rather than claiming methodological novelty for stacking or SHAP individually. The study has several limitations: it uses ten application-time features from a single lending platform and period, includes only loans with resolved outcomes, and assesses explanation stability through associative feature-importance patterns rather than causal relationships. Future work should validate the framework on additional lending datasets and periods, investigate broader feature sets and changing economic conditions, and assess whether the observed calibration and explanation-stability patterns persist under external validation.

VII. Future Work

Several extensions follow from this work. First, richer information could be used: the borrower

text fields and geographic detail excluded here, bureau-style variables, or macroeconomic indicators, together with a study of how added features change calibration and explanation stability. Second, inherently interpretable models such as generalized additive models or explainable boosting machines could be compared with post-hoc SHAP explanations, following the argument for interpretable-by-design models [27]. Third, stability could be built into training, for example through regularization or feature-selection strategies that penalize rank variation across resamples, and the stacking ensemble could be explained with model-agnostic estimators. Fourth, calibration under drift deserves attention: recalibration schedules, drift detection and online updating [29] could be evaluated against the fixed validation-period calibrator used here. Fifth, cost-sensitive thresholds and profit-based evaluation would link the metrics to lending decisions, and fairness audits of both predictions and explanations would be needed before deployment. Finally, replication on bank-lending and other P2P datasets would show whether the observed relations among discrimination, calibration and explanation stability generalize.

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AUTHOR CONTRIBUTIONS

Adnan Qadir: Data collection and dataset identification.

Tabassum Abbas: Methodology development and research design.

Rana Mohsin: Implementation and development of the proposed approach.

Muhammad Tahir: Research supervision, academic guidance, manuscript preparation, and paper formatting. Muhammad Tahir also served as the corresponding author.

All authors contributed to the development and refinement of the manuscript and approved its final version.

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